

Finite-Sample Spectral-Edge Exclusion from Gaussian Correlator Sketches

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<https://github.com/lluiseiriksson/finite-sample-spectral-certificates>

August 2026

Abstract

Spectral gaps are often inferred from noisy Euclidean correlators by fitting a small number of exponentials or solving a generalized eigenvalue problem [1]. Such point estimates do not by themselves control the error incurred when the spectral filter is selected from the same data. We give a finite-sample test that instead asks whether an entire simultaneous confidence set is compatible with a proposed upper edge of the visible transfer spectrum. For Gaussian sketches of a block correlator, an extreme-singular-value inequality yields a Loewner confidence band for the unknown joint Gram matrix. Intersecting this band with a truncated Hausdorff localizer is one semidefinite feasibility problem. We derive its semidefinite alternative: every rejection is accompanied by a covariance-aware matrix witness, a negative separation margin, and checkable primal-independent residuals. The test has type-I error at most α at fixed truncation order and requires neither known covariance nor a Gaussian approximation to the quadratic moment features. A second theorem gives an explicit witness-dependent sample size for power at least $1 - \alpha$, and one rejection is replayed with exact rational arithmetic. Simulations in the interacting axial next-nearest-neighbour Ising chain compare the certificate to covariance-known ellipsoids, plug-in eigenvectors, fixed Chebyshev filters, and Ritz estimates. A 16,800-SDP production grid resolves sample-size and moment- degree transitions. We also document two negative results: Hotelling calibration undercovers the non-Gaussian quadratic features, and the Gaussian Loewner band undercovers variance-matched Student- t_5 primitives. The method certifies exclusion of overly large *visible* gaps; it does not turn non-rejection into a lower bound on a Hamiltonian mass gap.

1 Introduction

Let H be a Hamiltonian with ground energy E_0 , choose a Euclidean step $\tau > 0$, and write $T = \exp[-\tau(H - E_0)]$. Matrix-valued correlator moments have the form

$$B_k = \Psi^* T^k \Psi, \quad k = 0, 1, \dots, \quad (1)$$

where the columns of Ψ are ground-state-subtracted probe states. The largest point in the support visible to these probes is $\rho_{\text{vis}} = e^{-\tau \Delta_{\text{vis}}}$. If a probe misses the first excitation then Δ_{vis} may exceed the Hamiltonian gap; this visibility qualification is unavoidable.

Recent work has developed semidefinite bootstraps for exact Euclidean correlators [2], convergent lower-gap certificates for qubit Hamiltonians [3], and convergent certified upper bounds on the bulk gap [4]. Our question is complementary:

Given finitely many correlated random observations of a block correlator, can one reject a proposed visible gap with simultaneous, finite-sample error control while choosing the separating filter adaptively?

The distinction matters. A negative eigenvalue of a noisy localizer is a data-dependent observation, not yet a confidence statement. Entrywise error bars ignore correlation; a filter trained on the same sample invalidates a pointwise interval; and a Gaussian approximation to quadratic observables may be inaccurate precisely in the moderate- n regime where certification is most valuable.

Our contribution is a statistical-conic layer that can be placed in front of a truncated moment localizer. In the Gaussian-sketch model the primitive sketch vector is Gaussian, but the reported block moments are quadratic and Wishart-like. We construct a confidence set for the primitive covariance itself, then test its compatibility with the spectral null. The resulting test is nonasymptotic and covariance-free. Its dual supplies an explicit rejection witness rather than asking the reader to trust an “infeasible” solver label.

Scope. At finite moment degree, localizer positivity is a necessary condition for a proposed upper spectral edge. Its violation rigorously excludes that edge, but non-violation does not prove a gap. Consequently our output is a certified upper bound on a visible gap (or a rejection of an overly large claimed gap), not a proof of a positive mass gap. This boundary also distinguishes the present result from Hamiltonian-algebra hierarchies that can certify lower bounds [3].

2 Block moments and the upper-edge localizer

Assume for clarity that H and the probes admit a real representation; the complex proper-Gaussian case has different fourth-moment constants and is not claimed here. The spectral theorem gives a positive-semidefinite matrix-valued measure μ on $[0, 1]$ such that

$$B_k = \int_{[0,1]} x^k d\mu(x). \quad (2)$$

For a proposed edge $\theta \in (0, 1]$ and degree N , define

$$\mathcal{L}_{N,\theta}(B) = [\theta B_{i+j} - B_{i+j+1}]_{i,j=0}^N. \quad (3)$$

If $\text{supp } \mu \subseteq [0, \theta]$, then for every vector polynomial $p(x) = \sum_{i=0}^N x^i v_i$,

$$\sum_{i,j} v_i^* (\theta B_{i+j} - B_{i+j+1}) v_j = \int (\theta - x) p(x)^* d\mu(x) p(x) \geq 0. \quad (4)$$

Thus $\mathcal{L}_{N,\theta}(B) \succeq 0$ is a necessary finite-window null. A violation says that the probes see spectral weight above θ , hence

$$\Delta_{\text{vis}} < -\tau^{-1} \log \theta. \quad (5)$$

This elementary localizer is not presented as novel; truncated moment and localization techniques have a long history [5, 6]. The new issue below is how to test it after observing a correlated random Gram matrix.

3 Gaussian sketches and a covariance confidence band

Put $u_{k,a} = T^{k/2} \psi_a$ for time index $k = 0, \dots, 2N + 1$ and channel $a = 1, \dots, c$. Draw independent standard real Gaussian vectors g_r and observe

$$z_{r,k,a} = \langle g_r, u_{k,a} \rangle. \quad (6)$$

After stacking (k, a) into one index, $z_r \sim \mathcal{N}(0, K)$ with

$$K_{(k,a),(\ell,b)} = \langle u_{k,a}, u_{\ell,b} \rangle = B_{(k+\ell)/2}[a, b]. \quad (7)$$

In particular, the k th diagonal time block of K is B_k . Although $z_{r,k}z_{r,k}^\top$ is an unbiased rank-one estimator of B_k , these quadratic features are not jointly Gaussian.

Let

$$S = \frac{1}{n} \sum_{r=1}^n z_r z_r^\top, \quad d = (2N + 2)c. \quad (8)$$

The following confidence set acts on all times and channels simultaneously.

Theorem 1 (Wishart–Loewner confidence band). *Let $z_1, \dots, z_n$ be independent $\mathcal{N}(0, K)$ vectors in $\mathbb{R}^d$, where $K \succeq 0$ is unknown and the zero mean is known. For $0 < \alpha < 1$, set*

$$\eta = \frac{\sqrt{d} + \sqrt{2\log(2/\alpha)}}{\sqrt{n}} < 1, \quad a = (1 + \eta)^{-2}, \quad b = (1 - \eta)^{-2}. \quad (9)$$

Then

$$\mathbb{P}\{aS \preceq K \preceq bS\} \geq 1 - \alpha. \quad (10)$$

The statement remains valid for singular K .

Proof. Write the $n \times d$ data matrix as $Z = GK^{1/2}$, with G standard Gaussian. The Gaussian extreme-singular-value inequality gives, with failure probability at most $2e^{-t^2/2}$,

$$\sqrt{n} - \sqrt{d} - t \leq s_{\min}(G) \leq s_{\max}(G) \leq \sqrt{n} + \sqrt{d} + t.$$

With $t = \sqrt{2\log(2/\alpha)}$, division by $\sqrt{n}$, squaring, and congruence by $K^{1/2}$ yield $(1 - \eta)^2 K \preceq S \preceq (1 + \eta)^2 K$. Rearrangement gives (10). Congruence does not require invertibility of K , so the same argument covers the singular case. We use the ambient d rather than an unknown effective rank. See [7, 8] for the nonasymptotic singular-value bounds. $\square$

Remark 2 (Known mean is substantive). *If a sample mean is first estimated and subtracted, the degrees of freedom and constants change. The present theorem applies to the centered-by-construction sketches in (6); it must not be copied unchanged to an unknown-mean observation model.*

4 Exact compatibility test and semidefinite alternative

Let $P(K) = (K_{kk})_{k=0}^{2N+1}$ extract the diagonal time blocks. Define the random confidence set

$$\mathcal{C}_\alpha(S) = \{Q \in \mathbb{S}^d : aS \preceq Q \preceq bS\}. \quad (11)$$

We reject the proposed edge θ precisely when

$$\text{there is no } Q \in \mathcal{C}_\alpha(S) \text{ with } \mathcal{L}_{N,\theta}(P(Q)) \succeq 0. \quad (12)$$

This is one SDP feasibility problem.

Corollary 3 (Finite-sample validity). *Under the null $\mathcal{L}_{N,\theta}(P(K)) \succeq 0$, the probability that (12) rejects is at most α .*

Proof. On the event in theorem 1, the true K is feasible in (12); rejection is therefore impossible. The rejection event is contained in the confidence-set failure event. $\square$

No sample splitting is needed: arbitrary adaptive optimization inside the confidence set preserves this argument. This is the central statistical advantage over selecting a polynomial direction from S and attaching an unadjusted pointwise error bar.

The same event also gives a nonasymptotic power condition. It makes explicit how statistical dimension, spectral violation, and sketch count trade off.

Theorem 4 (Finite-sample detection condition). *Fix an alternative covariance $K \succeq 0$ and let $C_X = (\mathcal{L}_{N,\theta} \circ P)^*(X)$ for some $X \succeq 0$ with $\text{tr } X = 1$. Suppose*

$$\gamma_X = -\langle C_X, K \rangle > 0, \quad w_X = \text{tr} \left| K^{1/2} C_X K^{1/2} \right|. \quad (13)$$

If

$$\delta_+(\eta) w_X < \gamma_X, \quad \delta_+(\eta) = \left(\frac{1+\eta}{1-\eta} \right)^2 - 1, \quad (14)$$

then the test rejects with probability at least $1 - \alpha$. In particular, writing $r_X = \gamma_X / w_X$ (with the conclusion immediate if $w_X = 0$), the explicit sufficient sample size is

$$n > \left(\sqrt{d} + \sqrt{2 \log(2/\alpha)} \right)^2 \frac{(\sqrt{1+r_X} + 1)^4}{r_X^2}. \quad (15)$$

Proof. On the event used in Theorem 1, every $Q \in \mathcal{C}_\alpha(S)$ obeys

$$r_- K \preceq Q \preceq r_+ K, \quad r_\pm = \left(\frac{1 \pm \eta}{1 \mp \eta} \right)^2.$$

Thus, for $E = Q - K$, $-(1 - r_-)K \preceq E \preceq (r_+ - 1)K$. Compressing to $\text{ran } K$ and applying congruence by $K^{\dagger/2}$ shows

$$\langle C_X, E \rangle \leq (r_+ - 1) \text{tr}[(K^{1/2} C_X K^{1/2})_+] \quad (16)$$

$$+ (1 - r_-) \text{tr}[(K^{1/2} C_X K^{1/2})_-] \leq \delta_+(\eta) w_X. \quad (17)$$

Condition (14) therefore gives $\langle X, \mathcal{L}_{N,\theta}(P(Q)) \rangle = \langle C_X, Q \rangle < 0$ for every Q in the confidence set. No such Q can have a PSD localizer, so rejection occurs on an event of probability at least $1 - \alpha$. Finally, $\delta_+(\eta) < r_X$ is equivalent to $\eta < r_X / (\sqrt{1+r_X} + 1)^2$; substituting (9) gives (15). $\square$

The bound is deliberately witness-specific and conservative. It is useful because it is a genuine power guarantee, not because it predicts the sharp transition of an optimized SDP. Optimizing X for empirical power remains fully protected by Corollary 3; using X in Theorem 4 as a population design witness is only an analysis device.

For numerical auditability, let D be any fixed positive diagonal scaling computed from S , put $Q = D\tilde{Q}D$, and denote the scaled interval by $A \preceq \tilde{Q} \preceq B$. Let $\mathcal{L}_D$ be the resulting affine linear map.

Proposition 5 (Semidefinite alternative). *Assume $S \succ 0$. The feasibility problem (12) is infeasible if and only if there exist $Y_A, Y_B, Y_L \succeq 0$ such that*

$$Y_A - Y_B + \mathcal{L}_D^*(Y_L) = 0, \quad (18)$$

$$-\langle Y_A, A \rangle + \langle Y_B, B \rangle < 0. \quad (19)$$

The multipliers may be normalized by $\text{tr } Y_A + \text{tr } Y_B + \text{tr } Y_L = 1$.

Proof. Pair the three primal inequalities $\tilde{Q} - A \succeq 0$, $B - \tilde{Q} \succeq 0$, and $\mathcal{L}_D(\tilde{Q}) \succeq 0$ with the three PSD multipliers. Equation (18) cancels every term containing $\tilde{Q}$; primal feasibility would then force the constant in (19) to be nonnegative, proving one direction. Conversely, the interval is compact and has the strict midpoint $(A + B)/2$ when $S \succ 0$. Strong conic separation therefore supplies the stated multipliers whenever its intersection with the inverse image of the PSD localizer cone is empty. Homogeneity permits trace normalization. If S is singular, the same claim requires first restricting all matrices to $\text{ran } S$; we do not silently replace this reduction by a numerical ridge. $\square$

The implementation minimizes the dual margin under trace normalization. A reported rejection stores the margin, the Frobenius residual of (18), normalization error, and the smallest eigenvalue of each numerical multiplier. The production grid uses a conservative a-posteriori residual allowance. In addition, one deliberately wider confidence interval is rejected by an exact-rational witness whose PSD and strict-negativity checks use no numerical linear-algebra library; see Section 7.1.

5 Why covariance awareness changes the witness

The noiseless localizer test uses a minimum eigenvector. Under simultaneous uncertainty, that direction need not be optimal. For comparison, if a moment vector y has known mean covariance C/n and an ellipsoidal radius q , a trace-one PSD witness X has population signal-to-noise ratio

$$\kappa(X) = \frac{[-\operatorname{tr}(\mathcal{L}(y)X)]_+}{\sqrt{\ell(X)^\top C \ell(X)}}, \quad \ell(X)_k = \operatorname{tr}(L_k X). \quad (20)$$

The robust upper bound becomes negative exactly when $\sqrt{n} \kappa(X) > q$. Optimizing X therefore trades localizer negativity against covariance sensitivity, rather than simply minimizing the observed eigenvalue.

Mixed witnesses can be strictly better in principle. For the synthetic family $M = -\varepsilon I_r$ with isotropic Frobenius uncertainty, every rank-one projector has uncertainty norm one, whereas $X = I_r/r$ has norm $r^{-1/2}$. The sample threshold differs by a factor r . In our ANNNI pilots, however, many optimized ellipsoidal witnesses above threshold are numerically rank one. The observed advantage there is primarily *covariance-aware direction selection* relative to the plug-in minimum eigenvector, not evidence that a high-rank mixture is always required.

6 Interacting ANNNI experiment

We use the periodic axial next-nearest-neighbour Ising chain

$$H = -J_1 \sum_i Z_i Z_{i+1} - J_2 \sum_i Z_i Z_{i+2} - h_x \sum_i X_i \quad (21)$$

with $J_1 = 1$, $J_2 = 0.37$, $h_x = 2.2$, and $\tau = 0.2$. This frustrated, nonintegrable finite chain [9] is not being used to claim a new phase boundary; it is a controlled interacting test bed where exact sparse diagonalization and Krylov propagation provide ground truth. Ground-state-subtracted central-site X and Z probes form a two-channel correlator. The global spin-flip parity separates their visible sectors.

At degree $N = 2$, the primitive joint sketch dimension is $d = 12$, the flattened quadratic moment dimension is 18, and the localizer has size 6×6 . Lengths $L = 6, 8, 10, 12, 14, 16$ are tested at the exact visible edge and at gaps inflated by 15% and 35%. For each cell we generate independent sketch ensembles at $n = 500, 1000, 2000, 5000$. The production seed, full raw summaries, and SHA-256 hashes are stored in the repository.

6.1 Baselines and controls

We compare four logically different objects.

1. The Wishart–Loewner dual certificate of Proposition 5, with unknown covariance and finite-sample level control.
2. A covariance-known multivariate-Chebyshev ellipsoid, useful as a controlled oracle reference but not a deployable unknown-covariance method.

3. A plug-in minimum-eigenvector and a fixed Chebyshev polynomial filter. These expose the value of covariance-aware direction selection.
4. A generalized Ritz/GEVP edge estimate. It is a point estimator, not a confidence certificate, and is reported separately rather than granted the same logical status.

The known-covariance pilot showed a sharp sample transition: for the 35% inflated gap, the covariance-aware robust test reached essentially full power near $n = 1000$ across $L \leq 14$, while the plug-in eigenvector was usually still silent until $n = 5000$. At the true edge the robust methods made no rejection in the medium grid. These observations motivated the covariance-unknown production campaign.

6.2 Production power and moment-degree transition

The main unknown-covariance campaign comprises 7,200 independently generated SDPs. It made zero rejections in 2,400 null ensembles and 2,400 rejections in 2,400 ensembles at 35% inflation. For the harder 15% alternative, the finite-size transition is resolved in Table 1. Each entry is the number of rejections among 100 ensembles; Wilson intervals are plotted in Figure 1. Across all 7,200 solves, the largest conservative stationarity allowance was 3.76×10^{-13} and the smallest reported eigenvalue of a shifted multiplier was 1.40×10^{-14} .

Table 1: Safe rejections out of 100 for a 15% inflated visible-gap claim at moment degree $N = 2$.

L	$n = 500$	$n = 1000$	$n = 2000$	$n = 5000$
6	100	100	100	100
8	100	100	100	100
10	98	100	100	100
12	51	100	100	100
14	0	86	100	100
16	0	7	97	100

The independent 7,200-SDP degree ablation made zero null rejections in 3,600 ensembles. Degree one is powerless at $L = 12, 16$ throughout the tested range; degree three reaches unit power in every $L = 8, 12$ cell and at $n \geq 1000$ for $L = 16$. This is not the tautology that adding moments can only help: the Wishart dimension penalty grows with N . The observed dominance therefore quantifies a favorable approximation–concentration tradeoff for this model.

6.3 Conservatism of the analytic band

In 10,000 standard-Gaussian calibration replicates at $d = 12$, the analytic η values were 0.276, 0.195, 0.138 for $n = 500, 1000, 2000$, whereas the empirical 95th percentiles of the required singular-value deviation were 0.173, 0.122, 0.0862. The analytic width was consistently about 1.60 times the empirical quantile and covered all 10,000 replicates in each cell. Thus the strong production power is not obtained by tuning the band to its Monte Carlo distribution.

6.4 Negative calibration result

The quadratic sketch features obey Isserlis’ covariance formula but are Wishart-like rather than multivariate Gaussian. At $L = 12$, $N = 2$, and only $n = 120$ sketches, a nominal 95% Hotelling ellipsoid covered the exact moment vector in 83.4% of 500 simulated ensembles. Its apparent 97.4% detection rate for an inflated gap was therefore invalid as a 95% certificate. Coverage approached nominal only slowly with n . This failed route is retained in the artifact as a warning against importing an exact Gaussian-mean theorem into a quadratic-feature problem.

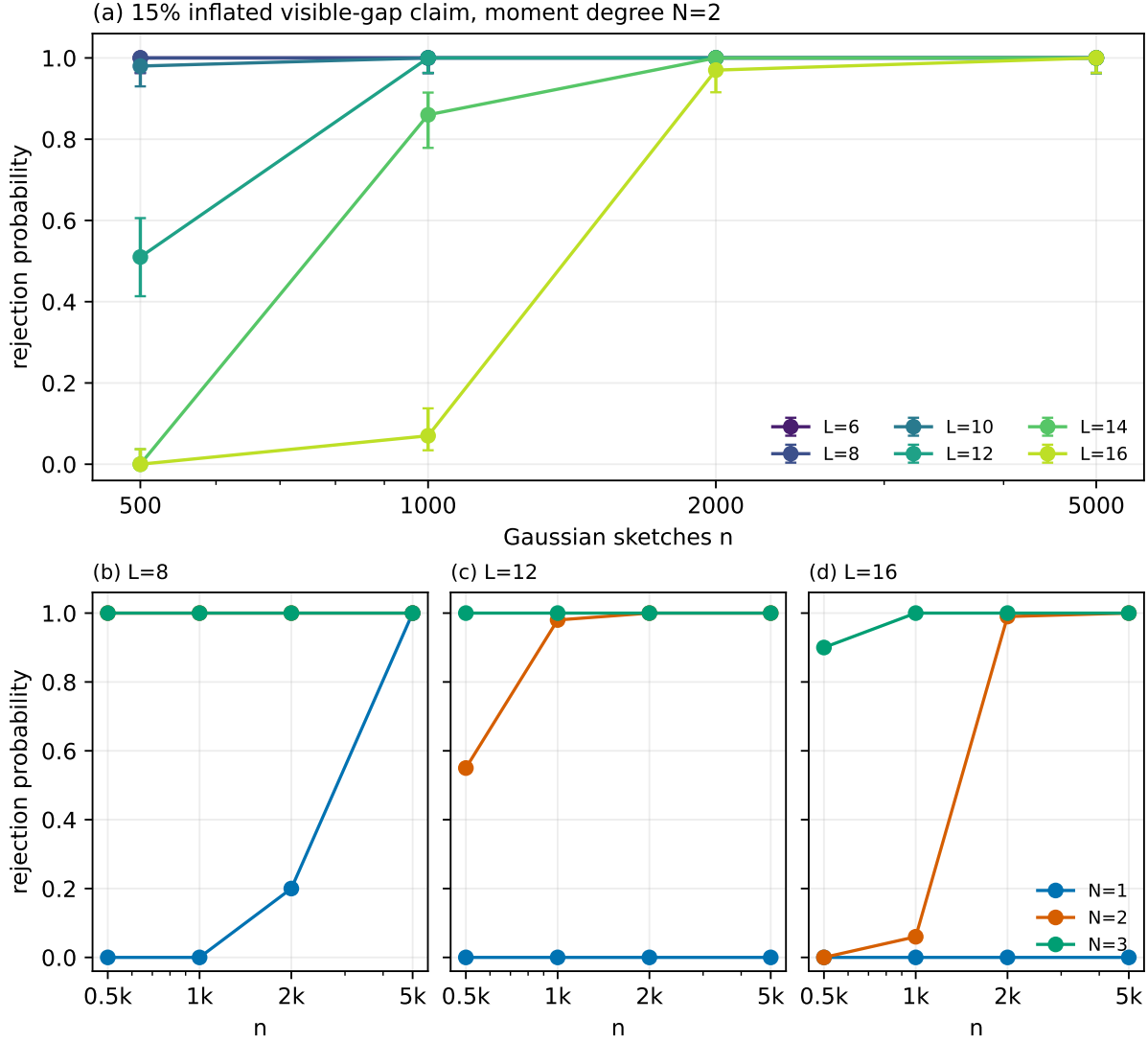


Figure 1: Finite-sample power. (a) The $N = 2$ production campaign with 95% Wilson intervals. (b–d) Independent degree ablation at 15% inflation. Increasing N raises the primitive dimension from 8 to 16 and widens the Loewner band, yet the stronger polynomial localizer dominates: at $L = 16$ and $n = 500$, power rises from 0 for $N = 1, 2$ to 0.90 for $N = 3$.

6.5 Primitive-law misspecification

The Wishart–Loewner theorem is exact for Gaussian primitive sketches, not for all zero-mean isotropic laws. We tested this boundary in 2,400 further SDPs, using Gaussian, variance-matched Student- t_5 , and Rademacher primitive coordinates before applying the same population square root. The Gaussian band covered in all 800 in-model ensembles. Rademacher sketches also happened to cover in all 800 tested ensembles, but this is empirical conservatism, not a theorem. Student- t_5 coverage was only 0.780 at $n = 500$ and 0.738 at $n = 1000$ after pooling the two independently generated gap-factor cells; its heavy radial tails do not become licensed merely by covariance matching.

7 Numerical verification protocol

Ill-conditioned Euclidean moment matrices make raw primal status unreliable. In a first primal–dual Wishart replay, two of five inflated-gap samples were labeled `optimal_inaccurate` by a

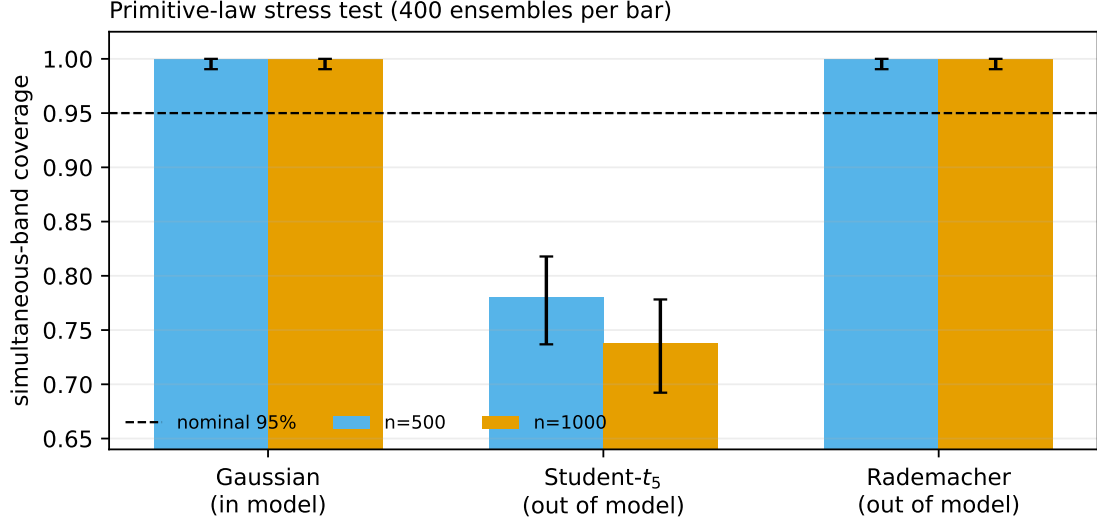


Figure 2: Coverage stress test for the simultaneous Loewner band, with Wilson intervals. Each bar pools 400 ensembles. Only the Gaussian bars are covered by Theorem 1; the other two laws are deliberate negative controls.

primal solver even though their dual margins were below -4×10^{-4} . In a separate ten-ensemble $L = 8$ smoke at $n = 500$, null dual objectives lay between 1.8×10^{-10} and 6.5×10^{-10} , while 35%-inflated-gap objectives lay between -8.2×10^{-4} and -4.1×10^{-4} . The maximum stationarity residual was 2.3×10^{-9} . The first production pass conservatively thresholded the raw dual margin at -10^{-7} and stored all residual diagnostics. The final pass uses the feasibility repair described next.

Feasibility repair. Let floating-point multipliers first be projected onto the PSD cone, and let $R = Y_A - Y_B + \mathcal{L}_D^*(Y_L) = R_+ - R_-$ be their remaining stationarity residual. The replacements

$$\hat{Y}_A = Y_A + R_-, \quad \hat{Y}_B = Y_B + R_+ \quad (22)$$

remain PSD and satisfy stationarity exactly, up to the arithmetic used for the eigendecomposition. Trace renormalization preserves this equality. In a 20-ensemble repaired smoke, the maximum stationarity residual fell below 8×10^{-17} and normalization error below 4×10^{-16} ; null objectives were positive, whereas all 15% and 35% inflated-gap objectives were negative. The production threshold is -10^{-9} on the *repaired* objective, leaving an additional buffer above double-precision roundoff.

For an additional a-posteriori check, let $\hat{R}$ be the final residual and recall that every scaled primal matrix obeys $0 \preceq \tilde{Q} \preceq B$. Then

$$|\langle \hat{R}, \tilde{Q} \rangle| \leq \|\hat{R}\|_* \|B\|_{\text{op}}. \quad (23)$$

We add this worst-case quantity to the repaired dual objective before making a decision. The code also shifts each multiplier by an explicit roundoff-scale identity so that its reported minimum eigenvalue is positive. The final rule therefore tests a conservative upper bound on the dual contradiction margin, not the raw solver objective.

This gap between scientific margin and numerical residual is encouraging but does not by itself constitute exact arithmetic. The release uses three increasing levels of assurance:

1. solve in floating point after diagonal congruence scaling;
2. independently rebuild the adjoint and residual from exported matrices;

3. repair PSD and stationarity defects while adding their worst-case effect to the contradiction margin;
4. replay one wider-band representative with exact rational arithmetic.

7.1 Exact-rational representative witness

For $L = 8$, $N = 2$, $n = 500$, and the rational proposed edge $\theta = 3/5$, we replace the analytic deviation by the larger rational value $\bar{\eta} = 3/10$. This widening is itself certified by elementary rational bounds: $\exp(15/4) > 40$ via a positive Taylor partial sum, $\sqrt{12} < 7/2$, $\sqrt{2 \log 40} < 11/4$, and $25/4 < 3\sqrt{5} = (3/10)\sqrt{500}$. Hence infeasibility for

$$\frac{100}{169}S \preceq K \preceq \frac{100}{49}S \quad (24)$$

implies infeasibility for the narrower 95% analytic band.

The stored Gaussian coordinates are hexadecimal IEEE-754 values. The replay interprets each as its exact dyadic rational, rebuilds S by rational dot products, and reads rounded dual multipliers with a common denominator 10^{14} . A diagonal shift of 10^{-11} makes all three multipliers strictly positive definite. Exact fraction LDL decompositions give a minimum pivot 2.85×10^{-11} . Instead of a floating nuclear norm, the replay uses the coarser exact inequality

$$|\langle R, Q \rangle| \leq d \max_{ij} |R_{ij}| \operatorname{tr} B, \quad 0 \preceq Q \preceq B. \quad (25)$$

After this allowance, the exact rational certificate upper bound is $-6.47697914332 \times 10^{-4}$. The JSON artifact has SHA-256 (line break is typographical)

```
aa9e6af05079ca823b392d2ee4889928
2b87d32839021bd84b6085bd5f8d842d
```

and is independently replayed by `verify_rational_witness.py`. This exact statement is conditional on the stored finite-precision sample; the Gaussian idealization supplies the statistical coverage statement, not the rational replay itself.

8 Limitations and interpretation

Visible versus Hamiltonian gap. The probes may have zero overlap with the first excitation. Multiple symmetry channels reduce but do not remove this risk. Every result is consequently labeled Δ_{vis} unless a separate overlap theorem is available.

Upper versus lower bounds. Rejecting support in $[0, \theta]$ proves that a claimed gap is too large. A finite localizer’s non-rejection is not a proof that the true support lies below θ . Our method therefore complements, rather than replaces, lower-gap Hamiltonian certificates [3].

Acquisition model. The exact confidence band uses independent, known-zero-mean Gaussian sketch coordinates. Classical stochastic estimators or randomized numerical linear algebra can realize this model directly; a laboratory implementation must state how those coordinates are obtained and account for state-preparation and measurement noise. Markov-chain autocorrelation, non-Gaussian primitive sketches, and unknown centering require different concentration results.

Finite volume. The ANNNI study validates statistical behavior at finite L . It neither establishes a thermodynamic gap nor competes with the convergent bulk hierarchy of Xu et al. [4]. A future combination of noisy-data confidence sets with thermodynamic operator-algebra hierarchies is an open direction.

9 Conclusion

The main result is a finite-sample bridge from random correlator sketches to an auditable spectral-edge exclusion. A single Loewner band absorbs all time-channel correlations and all adaptive witness selection; a semidefinite alternative turns rejection into a replayable matrix inequality. The approach is intentionally one-sided. It can falsify an overly large visible gap with controlled type-I error, but it does not infer a mass gap from silence. The interacting-chain campaign is designed to quantify when that rigorous statement is statistically useful and when conventional point estimates are merely more optimistic.

Reproducibility. Code, theorem audit notes, raw pilot outputs, the Colab production notebook, and the frozen production artifacts are available at <https://github.com/lluiseiriksson/finite-sample-spectral-certificates>. The three production JSON files have SHA-256 digests recorded in `results/production/summary.json`; the continuous-integration workflow rechecks the localizer adjoint, primal–dual smoke, safe Wishart smoke, and exact-rational replay.

Acknowledgements. OpenAI Codex assisted with code generation, numerical orchestration, literature triage, and adversarial consistency checks. The author retains responsibility for all claims and the final manuscript.

A Strong alternative from compact interval separation

We spell out the separation argument behind Proposition 5, since generic semidefinite systems can exhibit weak infeasibility. Let

$$\mathcal{D} = \{Q : A \preceq Q \preceq B\}, \quad \mathcal{M} = \{\mathcal{L}_D(Q) : Q \in \mathcal{D}\}.$$

When $A \prec B$, $\mathcal{D}$ is compact, convex, and has a relative-interior point; hence $\mathcal{M}$ is compact and convex. If $\mathcal{M} \cap \mathbb{S}_+ = \emptyset$, strong separation from the closed PSD cone provides a nonzero matrix $Y_L \succeq 0$ such that

$$\sup_{Q \in \mathcal{D}} \langle Y_L, \mathcal{L}_D(Q) \rangle < 0. \quad (26)$$

The separator must be PSD: otherwise its linear functional is unbounded below or above along a PSD ray, with the orientation chosen in (26).

The support problem on the left of (26) is an SDP with a strictly feasible interval point $(A + B)/2$. Its Lagrange dual therefore attains the same optimum. Introducing $Y_A, Y_B \succeq 0$ for $Q - A \succeq 0$ and $B - Q \succeq 0$ makes the coefficient of Q vanish exactly when

$$Y_A - Y_B + \mathcal{L}_D^*(Y_L) = 0,$$

and leaves the constant $-\langle Y_A, A \rangle + \langle Y_B, B \rangle$. Strict negativity follows from (26). This proves the nontrivial direction of the alternative without invoking a blanket “SDP Farkas lemma” in a regime where it might be false. Conversely, pairing any feasible Q with such multipliers gives an immediate contradiction, as in the main text.

If S is singular, then A and B have the same kernel and no ambient strict midpoint exists. Because $A \preceq Q \preceq B$ forces $\ker S \subseteq \ker Q$, one may compress every matrix to $\text{ran } S$ and repeat the argument there. The current production ensembles have $n > d$ and a full-rank population Gram; the explicit range reduction is nevertheless preferable to adding an undocumented ridge.

B Exact robust separation for ellipsoidal moment errors

The covariance-known comparator used in the pilot follows a more general finite-dimensional identity. Let $y \in \mathbb{R}^p$ parameterize an affine localizer

$$L(y) = L_0 + \sum_{k=1}^p y_k L_k$$

and let

$$\mathcal{E} = \{\hat{y} + Rz : \|z\|_2 \leq q\}$$

be any compact ellipsoid. For a trace-one $X \succeq 0$ define $\ell(X)_k = \text{tr}(L_k X)$ and

$$U_{\mathcal{E}}(X) = \text{tr}[L(\hat{y})X] + q\|R^{\top}\ell(X)\|_2. \quad (27)$$

Theorem 6 (Exact ellipsoidal incompatibility). *The intersection $\mathcal{E} \cap \{y : L(y) \succeq 0\}$ is empty if and only if*

$$\min_{X \succeq 0, \text{tr } X=1} U_{\mathcal{E}}(X) < 0. \quad (28)$$

Proof. The image $L(\mathcal{E})$ is compact and convex. If it is disjoint from the closed PSD cone, strong separation supplies a PSD separator X ; normalization to trace one only fixes its nonzero scale. The support function of the ellipsoid in that direction is exactly (27). Conversely, if (28) is negative, then $\text{tr}[L(y)X] < 0$ for every $y \in \mathcal{E}$, which is impossible when both $L(y)$ and X are PSD. $\square$

For a singleton ellipsoid, (28) reduces to $\lambda_{\min}(L(y))$ and admits a rank-one eigenvector witness. Under uncertainty, however, the optimizer minimizes a sum of signal and sensitivity. It may be higher rank, and even when it is rank one it need not be the observed minimum eigenvector. The same coverage-event argument as in Corollary 3 shows that optimizing X on the data does not inflate type-I error when $\mathcal{E}$ is a simultaneous confidence set.

C Quadratic-feature covariance and the Hotelling control

For completeness, Isserlis' identity gives the exact covariance of one quadratic sketch replicate. With joint covariance entries $K_{ka,\ell c} = \mathbb{E}[z_{k,a}z_{\ell,c}]$,

$$\text{Cov}(z_{k,a}z_{k,b}, z_{\ell,c}z_{\ell,d}) \quad (29)$$

$$= K_{ka,\ell c}K_{kb,\ell d} + K_{ka,\ell d}K_{kb,\ell c}. \quad (30)$$

For the mean of n independent sketches this matrix is divided by n . If it were known, multivariate Chebyshev would give a distribution-free ellipsoid with squared radius r/α on its rank- r range. That oracle route is exact but conservative and requires the unknown K through (30); it is used only as a controlled comparator.

By contrast, the standard one-sample Hotelling pivot is exact when the *feature vectors themselves* are multivariate Gaussian. Products of Gaussian coordinates violate that hypothesis. The 83.4% coverage observed at nominal 95% and $n = 120$ is therefore not a paradox; it is a direct stress test of a misapplied calibration. The Wishart–Loewner route avoids this error by constructing its pivot before the quadratic map.

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